The documentation and process conversion measures necessary to comply with this revision shall be completed by 15 July 1994

INCH-POUND

MIL-S-19500/225E
15 April 1994
SUPERSEDING
MIL-S-19500/225D
21 April 1967

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON,
TYPES 2N1711, 2N1711S, 2N1890, 2N1890S, JAN AND JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, low-power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3 (similar to TO-5).

1.3 Maximum ratings.

Type 1/	P_T 2/ T_C = +25°C	P_T 3/ T_A = +25°C	V_{CBO}	V_{EBO}	I_C	V_{CER} R_{BE} = 10 Ω	T_j and T_{stg}
	W	W	V dc	V dc	mA dc	V dc	°C
2N1711	3.0	0.8	75	7	500	50	-65 to +200
2N1890	3.0	0.8	100	7	500	80	-65 to +200

1/ Also applies to the corresponding "S" suffix device.
2/ Derate linearly at 17.2 mW/°C for T_C > +25°C.
3/ Derate linearly at 4.57 mW/°C for T_A > +25°C.

1.4 Primary electrical characteristics.

Limits	h_{FE1} 1/	h_{FE2} 1/	$\|h_{fe}\|$	$V_{CE(SAT)}$	
				2N1711 2/	2N1890 2/
	V_{CE} = 10 V dc I_C = 10 µA dc	V_{CE} = 10 V dc I_C = 150 mA dc	f = 20 MHz V_{CE} = 10 V dc I_C = 50 mA dc	I_C = 150 mA dc I_B = 50 mA dc	I_C = 50 mA dc I_B = 5.0 mA dc
				V dc	V dc
Min	20	100	3.5	0.2	0.2
Max		300	12	1.5	1.2

1/ Pulsed (see 4.5.1).
2/ Also applies to the corresponding "S" suffix device.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 8 (similar to TO-5), type T-1A for 2N1711 and 2N1890, and type T-1C for 2N1711S and 2N1890S.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the country of origin may be omitted from the body of the transistor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening (JANTX level). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement JANTX level only
1/	Thermal impedance (see 4.3.2)
9	Not applicable
11	h_{FE2}; I_{CBO1}
12	See 4.3.1
13	ΔI_{CBO1} = 100 percent of initial value or 5 nA dc; whichever is greater; Δh_{FE2} = ±15 percent of initial value;

1/ Thermal impedance shall be performed anytime after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this thermal test.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5).
P_T = 800 mW;
2N1711, 2N1711S: V_{CB} = 24 V dc.
2N1890, 2N1890S: V_{CB} = 48 V dc.

NOTE: No heat sink or forced air cooling of the devices shall be permitted.

4.3.2 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3131. The maximum limit and conditions for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of process control of actual measurements which characterizes the die attach process. When three lot date codes have exhibited control, the data from these three lots will be used to establish a fixed screening limit (not to exceed the group A limit). Once a fixed limit has been established, monitor all future sealing lots using a sample from each lot.

4.3.2.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3131 (read and record value $Z_{\theta JX}$) derived conditions, limits, and thermal response curve shall be supplied to the qualifying activity on the qualification lot prior to qualification approval.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$ in group A subgroup 2 inspection:

a. I_M measuring current . . . . . 10 mA.

b. I_H forward heating current . . . . . 10 A.

c. t_H heating time . . . . . 10 ms.

d. t_{MD} measurement delay time . . . . . 100 μs maximum.

The maximum limit for $Z_{\theta JX}$ in group A, subgroup 2 is $Z_{\theta JX}$ (maximum) = 4.5°C/W.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
B3	1027	T_A = room ambient as defined in the general requirements of of MIL-STD-750 (see 4.5); P_T = 800 mW dc; V_{CB} = 24 V dc for 2N1711 and 2N1711S; V_{CB} = 48 V dc for 2N1890 and 2N1890S.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Lead fatigue: Test condition E.
C6	1026	T_A = room ambient as defined in the general requirements of of MIL-STD-750 (see 4.5). P_T = 800 mW; V_{CB} = 24 V dc for 2N1711 and 2N1711S; V_{CB} = 48 V dc for 2N1890 and 2N1890S.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
E1	1051	Condition C, 500 cycles; 22 devices, c = 0.
E1	1071	
E2	1038	80 to 85% of rated V_R, T_A = 150°C, t = 1,000 hours
E3	----	Not Applicable.
E4	3131	See 4.5.2, 22 devices, c = 0.
E5	----	Not Applicable.

4.5 Method of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

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Used in tables I, II, and III) are combined in the subgroup.

TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.4.1	$Z_{\theta JX}$		4.5	°C/W
Breakdown voltage collector to base	3001	Bias condition D, I_C = 100 µA dc	$V_{(BR)CBO}$			
2N1711, 2N1711S				75		V dc
2N1890, 2N1890S				100		V dc
Breakdown voltage emitter to base	3026	Bias condition D, I_E = 100 µA dc	$V_{(BR)EBO}$	7		V dc
Breakdown voltage collector to emitter	3011	Bias condition D, I_C = 30 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N1711, 2N1711S				30		V dc
2N1890, 2N1890S				60		V dc
Breakdown voltage collector to emitter	3011	Bias condition D, I_C = 100 mA dc, pulsed (see 4.5.1), R_{BE} = 10 Ω	$V_{(BR)CER}$			
2N1711, 2N1711S				50		V dc
2N1890, 2N1890S				80		V dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}			
2N1711, 2N1711S		V_{CB} = 60 V dc			10	nA dc
2N1890, 2N1890S		V_{CB} = 80 V dc			10	nA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 5 V dc	I_{EBO}		5.0	nA dc
Collector to emitter saturated voltage	3071	I_C = 150 mA dc, I_B = 15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$			
2N1711, 2N1711S				0.2	1.5	V dc
2N1890, 2N1890S					5.0	V dc
Collector to emitter saturated voltage	3071	I_C = 50 mA dc, I_B = 5 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$			
2N1890, 2N1890S				0.2	1.2	V dc
Base emitter saturated voltage	3066	Test condition A, I_C = 150 mA dc, I_B = 15 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.3	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Base emitter saturated voltage	3066	Test condition A, I_C = 50 mA dc, I_B = 5 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$			
2N1890, 2N1890S					9.0	V dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 10 µA dc, pulsed (see 4.5.1)	h_{FE1}	20		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 150 mA dc, pulsed (see 4.5.1)	h_{FE2}	100	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 500 mA dc, pulsed (see 4.5.1)	h_{FE3}			
2N1711, 2N1711S				50		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D	I_{CBO2}			
2N1711, 2N1711S		V_{CB} = 60 V dc			10	µA dc
2N1890, 2N1890S		V_{CB} = 80 V dc			15	µA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	h_{FE1}	35		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 5 V dc, I_C = 1 mA dc	h_{fe}	80	200	
		V_{CE} = 10 V dc, I_C = 5 mA dc	h_{fe}	90	270	
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 50 mA dc, f = 20 MHz	$\|h_{fe}\|$	3.5	12	

See footnotes at end of table.

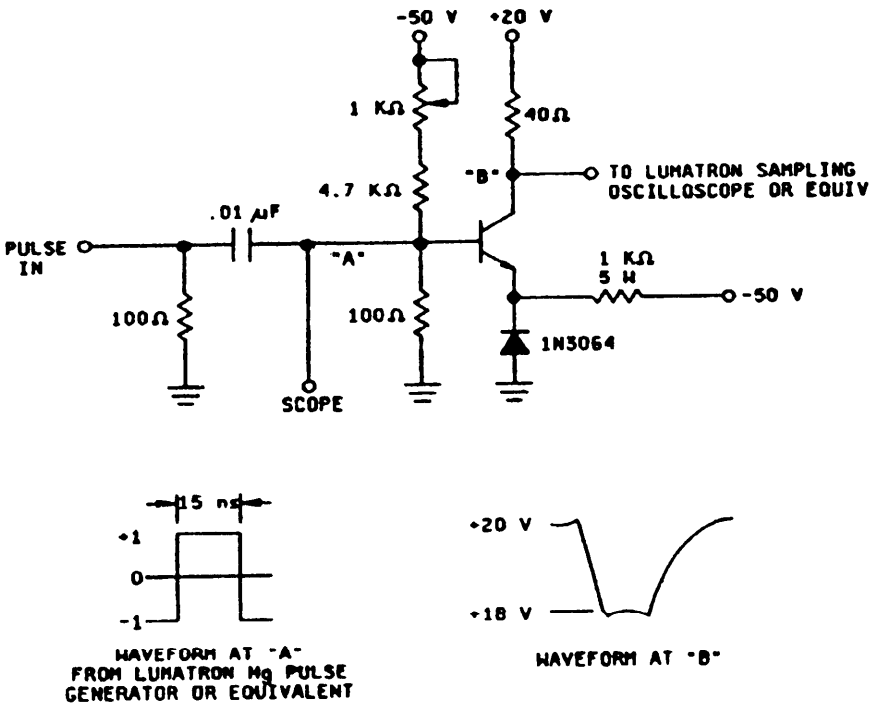
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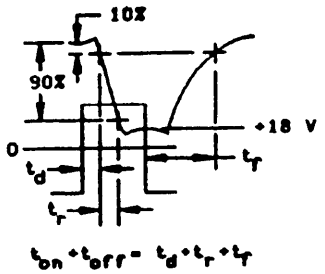
TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Small-signal short-circuit input impedance	3201	V_{CB} = 10 V dc, I_C = 5 mA dc	h_{ib}	4	8	Ω
Small-signal short-circuit output admittance	3216	V_{CB} = 10 V dc, I_C = 5 mA dc	h_{ob}	0	0.5	μΩ
2N1711, 2N1711S				0.0	1.0	μmhos
2N1890, 2N1890S				0.0	0.3	μmhos
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CB} = 10 V dc, I_C = 5 mA dc	h_{rb}			
2N1711, 2N1711S					$5x10^{-4}$	μmhos
2N1890, 2N1890S					$1.5x10^{-4}$	μmhos
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}			
2N1711, 2N1711S				8	25	pF
2N1890, 2N1890S				5	15	pF
Pulse response	3251	Test condition A, except test circuit and pulse requirements See figure 1 herein.	$t_{on} + t_{off}$		30	ns
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.
2/ Need not be performed in screen 13.

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NOTES:
1. The rise time (T_r) of the applied pulse shall be 2.0 ≤ ns, duty cycle ≤ 2%, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: Z_{in} ≥ 100 KΩ, C_{in} ≤ 12 pF, rise time ≤ .2 ns.

FIGURE 1. Pulse response (turn-on plus turn-off) measurement circuit and waveforms.

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4.5.2 Thermal resistance (qualification only). Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be 28 mA dc (2N1711 and 2N1711S), and 47 mA dc (2N1890 and 2N1890S).

b. Collector to emitter voltage magnitude ≥ 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be 25°C ≤ T_R ≤ 75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit shall be $R_{\theta JC}$ = 58°C/W.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish (see 3.3.1).

c. Type designation and quality assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Navy - EC
Army - ER
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, CG, MC
Air Force - 13, 14, 19, 85, 99

Preparing activity:
DLA - ES

Agent:
DLA - ES

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	1. DOCUMENT NUMBER MIL-S-19500/225E	2. DOCUMENT DATE (YYMMDD) 94/04/15

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, TYPES 2N1711, 2N1711S, 2N1890, 2N1890S, JAN AND JANTX

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-253-6048 (2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center ATTN: DESC-ELDT Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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